Tutorial 4

Q 4: Show that $v(x) \equiv |x - x_0|^{2-n}$ is harmonic in $\mathbb{R}^n \setminus \{x_0\}$ for $n \geq 3$. Do the same for $v(x) \equiv \log|x - x_0|$ if $n = 2$.

Solution:

Let $|x - x_0| = r = \sqrt{(x_1 - x_0^1)^2 + \cdots + (x_n - x_0^n)^2}$

$$\frac{\partial r}{\partial x_i} = \frac{2\,(x_i - x_0^i)}{2\sqrt{(x_1 - x_0^1)^2 + \cdots + (x_n - x_0^n)^2}}$$

$$= \frac{(x_i - x_0^i)}{r}$$

$$v(x) = r^{2-n}$$

$$\frac{\partial v}{\partial x_i} = (2-n)\, r^{1-n} \frac{\partial r}{\partial x_i}$$

$$= (2-n)\, r^{-n} (x_i - x_0^i)$$

$$\frac{\partial^2 v}{\partial x_i^2} = (-n)(2-n)\, r^{-n-1} (x_i - x_0^i) \times \frac{\partial r}{\partial x_i}$$
$$+ (2-n)\, r^{-n} \cdot 1.$$

$$= (-n)(2-n)\, r^{-n-2} (x_i - x_0^i)^2$$
$$+ (2-n)\, r^{-n}.$$

$$\Delta v = \sum_{i=1}^{n} \frac{\partial^2 v}{\partial x_i^2}$$

$$= (-n)(2-n)\, r^{-n-2} \left[(x_1 - x_0^1)^2 + \cdots + (x_n - x_0^n)^2 \right]$$
$$+ n(2-n)\, r^{-n}$$

$$= (-n)(2-n)\, r^{-n-2}\, r^2 +$$
$$n(2-n)\, r^{-n}$$

$$= 0.$$

$n=2$ Case

$$v(x) = \log r$$

$$\frac{\partial v}{\partial x_i} = \frac{1}{r} \cdot \frac{\partial r}{\partial x_i} = \frac{1}{r} \cdot \frac{(x_i - x_0^i)}{r}$$

$$= \frac{(x_i - x_0^i)}{r^2}.$$

$$\frac{\partial^2 v}{\partial x_i^2} = (x_i - x_0^i) \cdot \frac{-2}{r^3} \cdot \frac{\partial r}{\partial x_i}$$

$$+ \frac{1}{r^2}.$$

$$= (x_i - x_0^i)^2 \times \frac{-2}{r^4} + \frac{1}{r^2}.$$

$$\Delta v = \sum_{i=1}^{2} \frac{\partial^2 v}{\partial x_i^2} = \frac{-2}{r^2} + \frac{2}{r^2}$$

$$= 0.$$